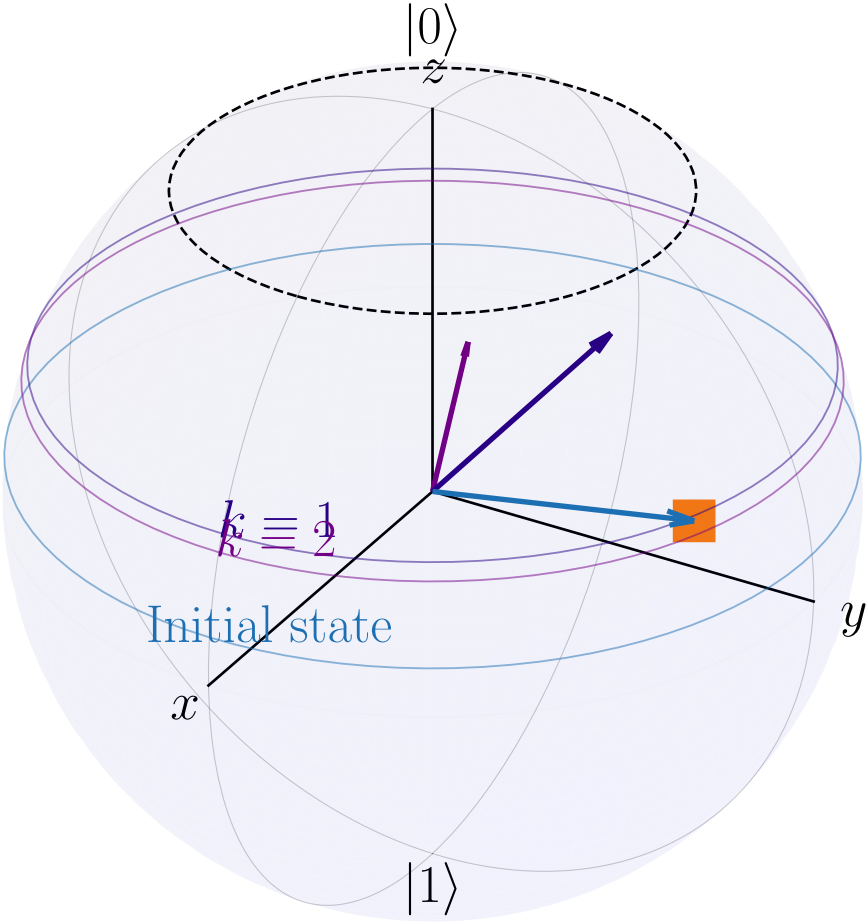

$|0\rangle$
z
$k \equiv \frac{1}{2}$
Initial state
y
x
$|1\rangle$